YUJIA TIAN

(+65)80675902 ◇ 13809260102@163.com ◇ FrundlesTian ◇ YuJia Tian

Educational Experiences

Nanyang Technological University (QS12) MSc in Software Project Management Jan 2026 - Jul 2027

Guangdong University of Foreign Studies B.Eng. in Software Engineering Sep 2021 - Jul 2025

Earned two national awards in the China Collegiate Computing Contest and five provincial awards in competitions including the "National Challenge Cup" and the College Students' Innovation and Entrepreneurship Program. Published four NLP papers as first author and was named an Outstanding Graduate.

Internship Experience

Shopee IP Singapore Private Limited - Shopee

Marketplace Intelligence & Data – Algo Data Aug 2026 - Present
Data Engineering Intern Singapore

Project Overview: Vimoo 1.0 is a released AIGC multimodal model, while **Vimoo-studio** focuses on building high-quality training data for AIGC models.
Tech Stack: Claude Code, Python, TypeScript, vLLM, Ray on Spark, HDFS
Key Contributions:

- Co-designed and developed Vimoo-studio's multimodal data-retrieval module, combining BM25 text matching with Faiss/Milvus vector search to provide recall services for video-generation training-data selection; iteratively improved retrieval efficiency and coverage.
- Participated in the online deployment, QPS benchmarking, and performance tuning of the internal Vimoo model with vLLM; built an offline batch-inference pipeline that generated captions for more than 5.7 million samples on B300/L40S clusters, supporting PT, CT, and SFT data delivery for video-generation models.
- Developing a unified video-frame service for data pipelines that decodes and caches frames once for reuse across processing stages, trading storage for compute to reduce redundant decoding overhead.

Shanghai Hanhui Information Technology Co., Ltd. - Temu

R&D Productivity Jul 2026 - Aug 2026
AI R&D Intern Shanghai, China

Project Overview: Cowizard is an internal coding-agent plugin designed for multiple business scenarios.
Tech Stack: Java, IntelliJ Platform, TypeScript, VS Code Extension API, WebSocket
Key Contributions:

- Led the migration of Cowizard Code Agent from IntelliJ-based IDEs to VS Code; improved the extension architecture and lifecycle management with TypeScript and the VS Code Extension API, designed a cross-IDE tool abstraction layer, and implemented streaming communication and automatic reconnection over WebSocket.

Apple R&D Shanghai Co., Ltd. - Apple

System In Packaging Feb 2025 - Aug 2025
Software Development Intern Shanghai, China

Project Overview: Perception is Apple's production-line test-data analytics platform, **CoEx** supports test-station validation and device-consistency analysis.
Tech Stack: Python, TypeScript, React, FastAPI, Polars, PySpark, Iceberg, Redis, JWT
Key Contributions:

- Designed the Runner background task system to mitigate blocking risks from long-running workloads such as Polars computation and large CSV parsing; implemented task persistence, retry, and failure recovery using a database queue, scheduler, and worker subprocesses.
- Migrated large-scale analytical data to a PySpark/Iceberg architecture to address table bloat caused by row-level BLOB storage in dataBinary; decoupled metadata retrieval from big-data reads, reducing lookup

latency from seconds to milliseconds.

- Developed authentication middleware based on AppleConnect+JWT with a two-layer fallback strategy; supported automatic JWT renewal and used Redis distributed locks to ensure mutual exclusion for concurrent updates.
- Built a React/FastAPI automation tool for CoEx to replace manual test-data querying, editing, and CSV preparation; supported batch anomaly generation, data pre-validation, and export of standardized test CSV files.

HSBC Global Service Centre - HSBC

Customer Service — Apr 2024 - Dec 2024
Algorithm Application Intern — Guangzhou, China

Project Overview: **Xiaofeng** is HSBC's ToC multi-layer decision-making intelligent customer service system, and **SQLhelper** is its NL2SQL auxiliary query tool for manual support workflows.
Tech Stack: RoBERTa, NL2SQL, Python, PHP
Key Contributions:

- Developed SQLhelper model training under resource constraints for customer-service retrieval demands; built an NL2SQL model based on RoBERTa with low-resource fine-tuning on financial domain corpora and optimized memory usage for lightweight inference deployment.
- Delivered SQLhelper engineering integration into Xiaofeng's manual support console; designed forbidden-syntax lists and SQL syntax validation to ensure safe query generation and lower the data retrieval burden for customer-service agents.

Project Experience

claude-code-telegram: Remote Interaction for Claude Code (★ 2.8k+ Stars) — Aug 2026
Tech Stack: Python, Claude Agent SDK, python-telegram-bot, AsyncIO, Pytest
Project Overview: A Telegram-based remote interface for Claude Code with session persistence, tool execution, and security auditing; contributed PR #217, which was merged into the main branch.
Key Contributions:

- Designed a Human-in-the-Loop approval flow for high-risk Bash, Write, and Edit tool calls, connecting Telegram Allow/Deny interactions with Claude SDK callbacks and prioritizing approval events to prevent message-queue deadlocks.
- Identified through end-to-end Bot/Claude testing that `allowed_tools` pre-authorization bypassed `can_use_tool`; dynamically removed controlled tools to enforce approval, with configurable timeout allow/deny behavior, fail-closed defaults, and full command/file-path display.

Skills

- **Programming Languages:** Proficient in Python and its asynchronous coroutine/GIL concurrency principles. Skilled in foundational backend languages Java/Go/Gin framework. Skilled in TS/React workflows.
- **Data Development:** Familiar with PostgreSQL and Redis, with deep understanding of caching and persistence principles. Skilled in Kafka/PySpark/Hive/Iceberg workflows.
- **Computer Fundamentals:** Familiar with computer networking principles and understanding of HTTP/TCP protocols. Familiar with operating system core concepts such as processes/threads, memory management, and file systems. Proficient in data structures and their variants. Proficient in writing web crawlers.
- **Engineering Tools:** Proficient in Docker, Git, Nginx, and CI/CD tools; skilled in service deployment, version collaboration, and production environment troubleshooting.
- **AI Infrastructure:** Experienced in model training, low-resource fine-tuning, lightweight inference deployment, model serving, and production integration of large-model applications.
- **AI/Agent:** Familiar with core Agent architectures such as ReAct and Plan-and-Solve; knowledgeable about Agent engineering concepts like Context Engineering and Harness Engineering with hands-on optimization experience; familiar with Tool Calling, RAG, MCP, Skills, and multi-agent systems.